

Nimish Adhikari
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Education

PhD in Biostatistics, Boston University School of Public Health **May 2026**

Department of Biostatistics, Boston University School of Public Health
Boston, MA

Dissertation: “Methods to account and correct for Index Trait Bias in studies of subsequent traits”

Advisor: Gina Peloso, PhD

MA in Biostatistics, Boston University School of Public Health **August 2025**

Department of Biostatistics, Boston University School of Public Health
Boston, MA

BS in Applied Mathematics and Computer Science **May 2020**

Tufts University, Boston, MA

Experience

Graduate Research Assistant, Boston University, Biostatistics Department, Sep 2020 – Present

- Developed Bayesian reweighting methods to correct for index trait bias arising in common-variant GWAS studies of subsequent traits using a Gaussian, spike-and-slab and Laplace prior for incident trait effect, which allows for accurate estimation of genetic effects affecting subsequent traits.
- Adapted existing Bayesian Mendelian Randomization (MR) methods, including Bayesian Weighted MR which uses a normal prior with a Bernoulli weighting factor and MR-HORSE, which uses a horseshoe prior to correct for index trait bias.
- Created genetic and individual level simulations with shared confounders and various values of genetic correlations in R to assess the performance of novel Bayesian reweighting and Bayesian MR methods in comparison to other non-Bayesian methods (traditional MR, Slopehunter, MR-EGGER) across realistic scenarios.
- Applied Bayesian MR methods to find bias corrected genetic associations for subsequent trait in large scale Electronic Health Record (EHR) data for incident and subsequent stroke, and publicly available summary statistics for BMI-adjusted fasting insulin from the GIANT consortium.
- Created genetic analysis and simulation pipelines using R, plink, REGENIE, bcftools, bedtools, and other bioinformatics software through bash scripting in a Linux/UNIX high performance cluster environment.

Biostatistician, Massachusetts Veterans Epidemiology Research and Information Collaborative (MAVERIC), Department of Veterans Affairs, Boston, Sep 2020-present

- Performed Genome Wide Association Studies (GWAS) which found novel genetic markers for disease phenotypes of initial and subsequent Stroke, Lymphedema, and Chronic Venous Insufficiency using data from the MVP cohort.
- Investigated the role of genetics in drug response for HbA1c after prescription of statins, oral diabetic agents and GLP-1 in the MVP cohort, using GWAS which is then compared and meta analyzed with results from similar analyses performed in FinnGen, to check for heterogeneity and find novel signals for SNPs.
- Created pipelines for analyses of GWAS using plink and REGENIE, and post-GWAS analyses in large-scale EHR datasets like the MVP and UK Biobank in different environments for analytical high-performance cluster computing environments available in the VA and the Department of Energy.
- Performed replication studies in the UK Biobank on the DNANexus platform, performing harmonization across UK Biobank and MVP platforms and performing GWAS and post GWAS analysis that included a variety of Mendelian Randomization methods in R and Polygenic Risk scoring using PRS-CS and PRS-CSx across biobank platforms,
- Developed and evaluated R package *clustra* and corresponding SAS macros that uses k-means clustering to cluster longitudinal EHR data with unequally spaced time intervals using blood pressure trajectories from MVP as a potential use case, with similar clustering performance between software, with application to blood pressure trajectories in VA EHR.
- Constructed definitions for patient-level clinical phenotypes using ICD codes, procedure codes and drug prescriptions using SQL to extract outcomes including blood pressure trajectories, COVID-19 testing and COVID-19 vaccinations from EHR data tables across the VA system.

Undergraduate Research Assistant, Mother Infant Research Institute, Tufts University, May 2017 – May 2020

- Extracted textural features, including grayscale histogram, Local Binary Pattern (LBP), and gray level cooccurrence matrix, from 2D microscopic and ultrasound images for mouse and human placenta.
- Created a classification algorithm using textural features in logistic regression modeling for placental calcification.

Publications

Journal Articles

- **Adhikari N**, Lunetta K, Gagnon D, Peloso G.M. 2026. “Bayesian Mendelian randomization methods for index trait bias correction in subsequent trait genome-wide association studies”, Human Genetics and Genomics Advances; 7
- Peloso, Gina M., **Adhikari N.**, et al. 2026. “Genome-Wide Association Study of Chronic Venous Insufficiency and Lymphedema in the Million Veteran Program.” Circulation: Genomic and Precision Medicine; 19-3
- Elmore, A.R.*, **Adhikari, N.***, Hartley, A.E., Aparicio, H.J., Posner, D.C., Hemani, G., Tilling, K., et al. 2024. “Protein Identification for Stroke Progression via Mendelian Randomization in Million Veteran Program and UK Biobank.”, Stroke, 55 (8): 2045–54.

- Dickerman, B.A., Gerlovin, H., Madenci, A.L., Muñiz, M.J.F., Wise, J.K., **Adhikari, N.**, Ferolito, B.R., et al. 2023. “Comparative Effectiveness of Third Doses of mRNA-Based COVID-19 Vaccines in US Veterans.”, *Nature Microbiology*, 8 (1): 55–63.
- Correia-Branco, A., Kallol, S., **Adhikari, N.**, Caiaffa, C.D., Jayaraman, N., Kashpur, O., Wallingford, M.C. 2020. “Placental Calcification: Long-Standing Questions and New Biomedical Research Directions.”, *Cardiovascular Calcification and Bone Mineralization*, 263–96.
- Nguyen, C. D., Correia-Branco, A., **Adhikari, N.**, Mercan, E., Mallidi, S., Wallingford, M.C. 2020. “New Frontiers in Placenta Tissue Imaging.”, *EMJ. Radiology*, 1 (1): 54.
- Peloso, G., **Adhikari, N.**, Young, M.M., Cho, K., Kinlay, S. (2026). “Genome-wide association study of Chronic Venous Insufficiency and Lymphedema in the Million Veterans Program” (*submitted*, *Circulation*)

Preprints

- **Adhikari, N.***, Gerlovin, H. *, Ostrouchov, G., Ehrbar, R., Dufour, A.B., Ferolito, B.R., Demissie, S., Costa, L., Ho, Y., Tarko, L., Begoli, E., Cho, K., Gagnon, D.R. (2025). “clustra: A multi-platform k-means clustering algorithm for analysis of longitudinal trajectories in large electronic health records data.” arXiv preprint arXiv:2507.00962.

Working Papers

- **Adhikari, N.**, Lunetta, K.L., Gagnon, D., Mohammed, S., & Peloso, G. (2026).” Exploring Bayesian methods for index trait bias in genome-wide association studies of subsequent traits” (*in progress*)

Teaching

- Teaching Assistant, BS 858, Statistical Genetics, Fall 2026
- Teaching Assistant, BS 852, Statistical Methods in Epidemiology, Spring 2023
- Teaching Assistant, BS 805, Intermediate Statistical Computing and Applied Regression Analysis, Fall 2023, Spring 2022
- Instructor, Statistical Bootcamp for Masters in Biostatistics program, Summer 2024

Posters and Presentations

- “Protein Identification for Stroke Progression via Mendelian Randomization in Million Veteran Program and UK Biobank”, CHARGE San Antonio, Texas, October 10, 2023. (Presentation)
- “Bayesian Mendelian Randomization methods for Index Trait Bias correction in subsequent trait genome-wide association studies,” STATGEN, Minneapolis, Minnesota, May 21, 2025. (Presentation)
- “Bayesian Shrinkage for index trait bias correction in subsequent trait studies”, ASHG, Boston, October 18, 2025. (Poster)

Memberships in Associations and Awards

- Statistical Reviewer, AHA Stroke, 2024 - 2025
- Member, Mu Sigma Rho, 2024

- Co-chair, Biostatistics Student Association, Boston University, 2021-2023

Relevant Skills

- R, SAS, Python, C++, C, Bash scripting, unix/linux, cluster computing, plink, REGENIE
- Relevant Classwork: Probability Theory, Hypothesis Testing, Statistical Genetics, Nonparametric Statistics, Linear Models, Analysis of Correlated Data, Intermediate Statistical Computing and Applied Regression, Statistical Methods in Epidemiology, Statistics in Clinical Trials